Exploratory analyses: relationships in the file drawer survey

This document is **hypothesis-generating**. It searches for potentially interesting relationships:

- between *demographic/career variables* (D*) and* variables of interest* (R*, I*, S*, H*, M*)
- among *variables of interest* (e.g., awareness → uptake; barriers → uptake)

Given the small sample size (N = 50) and many possible comparisons, treat results as *exploratory signals*, not confirmatory evidence.

Reading guide (how to interpret the output in this report)

- Many sections output a table with columns like `n`, `rho`, `p`, `p_adj`.
  - `rho` is a **Spearman rank correlation** (monotonic association) between one predictor and one outcome.
  - `n` is the number of respondents with **non-missing values in both variables** (complete-case for that pair).
  - `p` is the nominal p-value from `cor.test(..., method = "spearman")`.
  - `p_adj` is a **Benjamini–Hochberg (BH) false-discovery-rate adjustment** applied *within that table* (i.e., within that "screen" of many pairwise tests).
- When variables are binary (0/1) or counts, Spearman correlations are still defined; interpret them as rank/monotonic effects (not linear regression slopes).
- For any correlation, the report enforces `min_n = 10`; if fewer than 10 complete cases exist, the correlation is not computed.

Helpers (tables, plots, recoding)

This section defines helper functions used throughout the report:

- `recode_*()` functions convert Qualtrics-style string responses into numeric scales (e.g., S5 Likert → 1–5).
- `spearman_test()` computes one Spearman correlation for a given X–Y pair.
- `screen_spearman()` runs **many** Spearman correlations across all X–Y pairs you specify, then BH-adjusts the p-values within that screen.
- `plot_spearman()` draws a scatterplot with a fitted line and prints Spearman correlation in the panel.

Data + derived variables

What happens in this section

1) Load the public survey dataset `data_public/public_main.csv` into `results_raw`.
2) Define label dictionaries (e.g., what each `S5[SQ00x]` stands for) so later tables are readable.
3) Create a cleaned `results` data frame with:
   - numeric versions of ordinal items (e.g., S5, S6, H4, I5),
   - a small set of *derived indices* used repeatedly in EDA (e.g., attrition share, open-science uptake),
   - convenience demographic variables (e.g., `career_stage_ord`).

Key derived variables used below (plain-language definitions)

- **Demographics / productivity**
  - `years_active` = D3 (years active in experimental philosophy)

    - `methods_courses` = D5 (number of methods/statistics courses)
    - `projects_per_year` = D9, `pub_per_year` = D10, `quant_projects_per_year` = D11
    - `career_stage_ord` is an ordinal recode of D7 (Graduate student = 1 … Full professor = 5); `early_career = 1` if `career_stage_ord <= 3`
- **Attrition / file-drawer exposure (from R1 stage counts)**
    - `published_n` = R1[SQ006]
    - `attrition_pre_sub_n` = R1[SQ001] + R1[SQ002] + R1[SQ003] (data analysis + formal + professional presentation)
    - `attrition_post_sub_n` = R1[SQ004] + R1[SQ005] (submitted + reviewed manuscript)
    - `attrition_total_n` = `attrition_pre_sub_n + attrition_post_sub_n`
    - `attrition_share` = `attrition_total_n / (attrition_total_n + published_n)` when denominator > 0
- **Awareness / implications**
    - `I2_num` is extracted from the I2 response string (e.g., `"5 - Very familiar"` → 5)
    - `I4_familiarity_mean` is the mean of the 7 familiarity ratings (I4[…][1])
    - `I4_severity_mean` is the mean of the 7 severity ratings (I4[…][2])
- **Barriers / incentives / capacity**
    - `barriers_mean` = mean of 11 S5 items after recoding to 1–5 (`Not a reason at all` → 1 … `Extremely a reason` → 5)
    - `incentives_mean` = mean of 5 S6 items after recoding to 1–5
    - `training_need_mean` = mean of 5 H4 items after recoding to 1–5
- **Uptake of proposed solutions**
    - For each S4 item, `"Yes"` → 1, `"No"` → 0, `"I don't know"` → missing.
    - `open_science_present_prop` = proportion of S4 *present* items answered `"Yes"` among the non-missing S4 present items for that respondent.
    - `open_science_future_prop` = analogous proportion for S4 *future*.
    - `fd_practices_present_yes` / `fd_practices_future_yes` = number of `"Yes"` responses for a focused "file-drawer-targeted" subset (registered reports, replications, micropublications).
- **Data sharing / micropublications**
    - `data_shared_ever` is derived from H1: it equals 1 if a respondent selected *any* sharing route (H1[SQ001]–H1[SQ006] = Yes), and 0 if they selected H1[SQ007] ("No").
    - `micropublications_ord` recodes M1 into 0/1/2 for Never/Rarely/Sometimes.

Quick descriptive sanity check (key indices)

What this table is: a compact descriptive summary of all key variables/indices used in later EDA.
For binary variables (0/1), the mean is interpretable as the **proportion** of 1s.

label	n	M	SD
Years active in experimental philosophy	50	9.14	5.27
Published projects	50	6.98	9.66
Total attrition	50	4.16	5.31

label	n	M	SD
Willingness to change practices	50	4.08	0.85
Familiarity with the file drawer problem	46	4.04	1.05
Mean unacceptability of questionable research practices	50	3.97	0.59
Mean familiarity with consequences	50	3.91	0.90
Mean perceived severity of consequences	50	3.61	0.63
Attrition before submission	50	3.52	4.78
Mean motivating incentives	50	3.46	0.90
Mean training need for data sharing	50	3.01	1.05
Career stage (ordinal)	49	2.94	1.21

label	n	M	SD
Methods/statistics courses completed	50	2.90	3.52
Barrier subscale: time/incentives/pressure	50	2.86	1.21
Mean barriers to sharing/publishing	50	2.18	0.77
Barrier subscale: skills/technical barriers	50	2.14	0.92
Projects per year	50	1.95	1.76
Barrier subscale: external restrictions/policy	50	1.89	0.86
Quantitative projects per year	50	1.83	1.72
Institutional supports for open dissemination (count)	50	1.82	1.34
Peer-reviewed publications per year	50	1.80	1.62

label	n	M	SD
Barrier subscale: defensive concerns (IP/misuse/competition)	50	1.80	0.96
Future file-drawer-targeted practices (count)	50	1.46	1.03
Sponsor requirements for open dissemination (count)	50	1.38	1.40
Ever shared data	49	0.86	0.35
Future open-science uptake (proportion)	50	0.81	0.18
Early career (student/postdoc/assistant)	49	0.71	0.46
Current file-drawer-targeted practices (count)	50	0.66	0.82
Attrition at submission/review	50	0.64	1.27

label	n	M	SD
Current open-science uptake (proportion)	50	0.47	0.25
Attrition share = attrition/(attrition+published)	46	0.45	0.35
Micropublications frequency (ordinal)	50	0.22	0.51

Exploratory relationships (literature-motivated)

Common analysis template in Sections 1–4:
Each table is a *screen* of multiple pairwise associations computed with `screen_spearman()`:

- statistic: Spearman `rho`
- missingness: complete-case for each pair
- multiplicity: BH/FDR `p_adj` computed **within that table** (not across the whole document)

1) Resource constraints (S5) vs uptake (S4)

Goal: relate broad constraints and "context" variables to uptake behaviors.

- Predictors (`x_vars`):
  - `barriers_mean` (mean of 11 S5 barrier items; 1–5)
  - `sponsor_req_count` (count of sponsor requirements selected in S1; 0–4)
  - `institution_support_count` (count of institutional supports selected in S2; 0–5)
- Outcomes (`y_vars`):
  - uptake indices: `open_science_present_prop`, `open_science_future_prop`
  - file-drawer-targeted uptake: `fd_practices_present_yes`, `fd_practices_future_yes`
  - dissemination behaviors: `data_shared_ever`, `micropublications_ord`

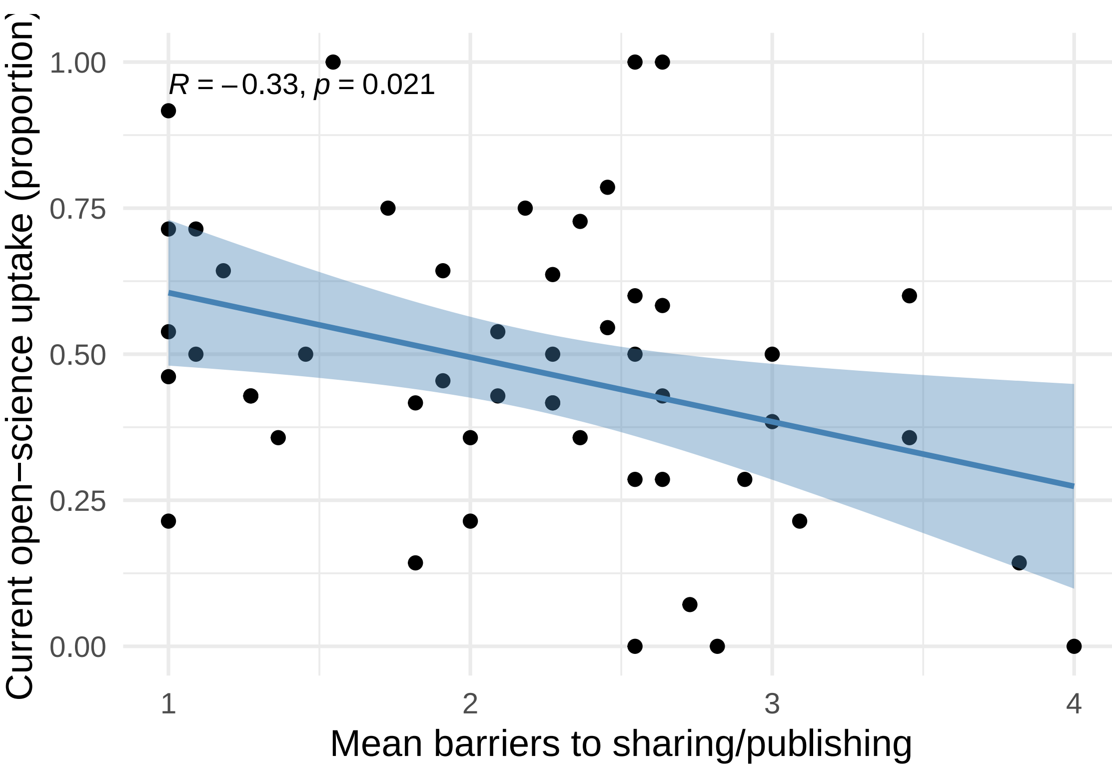
x_label	y_label	n	rho	p	p_adj
Mean barriers to sharing/publishing	Ever shared data	49	-0.39	0.01	0.10

x_label	y_label	n	rho	p	p_adj
Mean barriers to sharing/publishing	Current open-science uptake (proportion)	50	-0.33	0.02	0.19
Mean barriers to sharing/publishing	Future open-science uptake (proportion)	50	-0.28	0.05	0.28
Mean barriers to sharing/publishing	Micropublications frequency (ordinal)	50	-0.25	0.07	0.28
Sponsor requirements for open dissemination (count)	Current open-science uptake (proportion)	50	0.25	0.08	0.28
Sponsor requirements for open dissemination (count)	Future open-science uptake (proportion)	50	0.23	0.10	0.29
Sponsor requirements for open dissemination (count)	Ever shared data	49	0.23	0.11	0.29
Sponsor requirements for open dissemination (count)	Current file-drawer-targeted practices (count)	50	0.16	0.28	0.62

x_label	y_label	n	rho	p	p_adj
Mean barriers to sharing/publishing	Current file-drawer-targeted practices (count)	50	-0.15	0.31	0.62
Mean barriers to sharing/publishing	Future file-drawer-targeted practices (count)	50	-0.13	0.37	0.67
Sponsor requirements for open dissemination (count)	Micropublications frequency (ordinal)	50	0.12	0.41	0.67
Sponsor requirements for open dissemination (count)	Future file-drawer-targeted practices (count)	50	0.07	0.63	0.88
Institutional supports for open dissemination (count)	Current file-drawer-targeted practices (count)	50	0.07	0.64	0.88
Institutional supports for open dissemination (count)	Future open-science uptake (proportion)	50	-0.06	0.68	0.88
Institutional supports for open dissemination (count)	Ever shared data	49	0.04	0.78	0.90

x_label	y_label	n	rho	p	p_adj
Institutional supports for open dissemination (count)	Current open-science uptake (proportion)	50	-0.03	0.82	0.90
Institutional supports for open dissemination (count)	Micropublications frequency (ordinal)	50	0.02	0.89	0.90
Institutional supports for open dissemination (count)	Future file-drawer-targeted practices (count)	50	0.02	0.90	0.90

Visualization: scatterplot of `barriers_mean` vs `open_science_present_prop` with a linear smoother for readability; the annotation reports Spearman correlation.



1a) Which specific barriers relate to data sharing and uptake?

Goal: go from the overall `barriers_mean` index to the **individual S5 items**, to see which barrier types actually drive any associations.

- Predictors (`x_vars`): each of the 11 barrier items `S5[SQ001]`–`S5[SQ011]`, recoded to 1–5:
  - 1 = "Not a reason at all" … 5 = "Extremely a reason"
- Outcomes (`y_vars`):
  - `data_shared_ever` (0/1; derived from H1 as described above)
  - `open_science_present_prop` (0–1; proportion of S4-present items endorsed "Yes")
  - `attrition_share` (0–1; attrition/(attrition+published))

Each table below is a separate screen (separate BH correction within the table).

x_label	n	rho	p	p_adj
IP/ownership concerns	49	-0.58	0.00	0.00
Fear of misuse/misrepresentation	49	-0.49	0.00	0.00
Competitive advantage concerns	49	-0.40	0.00	0.02
Journal policies don't require sharing	49	-0.34	0.02	0.04
Fear of criticism/exposing errors	49	-0.34	0.02	0.04
Lack of training/skills	49	-0.30	0.04	0.07
Lack of direct benefits/incentives	49	-0.28	0.05	0.08
Technical challenges	49	-0.25	0.08	0.11

x_label	n	rho	p	p_adj
Pressure to publish in high-impact journals	49	-0.19	0.20	0.24
Extra effort feels over-whelming	49	-0.08	0.59	0.65
External restric-tions (fun-ders/part-ners)	49	0.02	0.91	0.91

Interpretation tip: for `data_shared_ever` (0/1), a negative `rho` means that higher barrier endorsement is associated with being in the "never shared" group.

x_label	n	rho	p	p_adj
Lack of train-ing/skills	50	-0.43	0.00	0.02
Fear of criti-cism/ex-posing errors	50	-0.39	0.01	0.03
Lack of direct bene-fits/in-centives	50	-0.37	0.01	0.03
Journal policies don't require sharing	50	-0.36	0.01	0.03
External restric-tions (fun-ders/part-ners)	50	0.35	0.01	0.03

x_label	n	rho	p	p_adj
Pressure to publish in high-impact journals	50	-0.23	0.10	0.19
Technical challenges	50	-0.22	0.13	0.20
Extra effort feels overwhelming	50	-0.21	0.14	0.20
Fear of misuse/misrepresentation	50	-0.05	0.71	0.87
IP/ownership concerns	50	-0.02	0.90	0.93
Competitive advantage concerns	50	-0.01	0.93	0.93

Here, `open_science_present_prop` is a respondent-level summary of "how many S4 practices do you currently do?".

x_label	n	rho	p	p_adj
External restrictions (funders/partners)	46	0.46	0.00	0.02
Fear of criticism/exposing errors	46	0.32	0.03	0.17
Pressure to publish in high-impact journals	46	0.29	0.05	0.17

x_label	n	rho	p	p_adj
IP/ownership concerns	46	0.27	0.07	0.17
Competitive advantage concerns	46	0.26	0.08	0.17
Lack of training/skills	46	0.25	0.09	0.17
Fear of misuse/misrepresentation	46	0.24	0.11	0.17
Journal policies don't require sharing	46	0.18	0.24	0.33
Extra effort feels overwhelming	46	0.16	0.30	0.33
Lack of direct benefits/incentives	46	0.16	0.30	0.33
Technical challenges	46	0.05	0.76	0.76

Here, `attrition_share` normalizes attrition by overall project volume (`attrition + published`) to reduce "more projects → more attrition" scale effects.

Reader-friendly group comparisons: ever vs never shared data

Because `data_shared_ever` is binary, it is often easier to read the results as **between-group differences** rather than as Spearman correlations with a 0/1 outcome.

In the table below we report group means/SDs and two common comparisons:

- **Welch t-test** (robust to unequal variances)
- **Mann–Whitney (Wilcoxon rank-sum)** as a nonparametric check (useful for ordinal Likert items)

`p_adj_*` are Benjamini–Hochberg adjustments computed **within this table** (across all variables listed).

Measure	n (never)	Never shared: M (SD)	n (ever)	Ever shared: M (SD)	ΔM (ever - never)	p_welch	p_adj_welch	p_wilcox	p_a
Current open-science uptake (proportion)	7	0.23 (0.13)	42	0.51 (0.24)	0.284	0.000	0.005	0.002	
Mean barriers to sharing/publishing	7	2.97 (0.74)	42	2.02 (0.67)	-0.955	0.013	0.035	0.007	
Barrier subscale: defensive concerns (IP/misuse/competition)	7	3.00 (1.02)	42	1.52 (0.61)	-1.476	0.008	0.032	0.000	
Barrier subscale: skills/technical barriers	7	2.93 (0.45)	42	2.04 (0.91)	-0.893	0.001	0.006	0.018	
Barrier subscale: time/incentives/pressure	7	3.48 (1.00)	42	2.74 (1.22)	-0.738	0.113	0.135	0.168	
Barrier subscale: external restrictions/policy	7	2.29 (0.95)	42	1.80 (0.83)	-0.488	0.238	0.260	0.151	
IP/ownership concerns	7	2.86 (1.21)	42	1.31 (0.60)	-1.548	0.015	0.035	0.000	
Fear of misuse/misrepresentation	7	3.00 (1.41)	42	1.40 (0.77)	-1.595	0.024	0.041	0.001	

Measure	n (never)	Never shared: M (SD)	n (ever)	Ever shared: M (SD)	ΔM (ever - never)	p_welch	p_adj_welch	p_wilcox	p_a
Competitive advantage concerns	7	3.14 (1.07)	42	1.86 (1.05)	-1.286	0.018	0.036	0.006	
Journal policies don't require sharing	7	3.00 (1.41)	42	1.86 (1.12)	-1.143	0.080	0.106	0.018	
Lack of training/skills	7	3.29 (1.11)	42	2.21 (1.26)	-1.071	0.047	0.070	0.041	
Extra effort feels overwhelming	7	3.71 (1.11)	42	3.29 (1.53)	-0.429	0.395	0.395	0.597	

2) Skills/training (D5, H4) vs uptake and barriers

Goal: test whether capacity/skills relate to uptake, barriers, and willingness to change.

- Predictors (`x_vars`):
  - `methods_courses` (D5; count of completed methods/statistics courses)
  - `training_need_mean` (mean of H4 training-need items; 1–5)
- Outcomes (`y_vars`):
  - uptake: `open_science_present_prop`, `fd_practices_present_yes`, `data_shared_ever`
  - constraints/motivation: `barriers_mean`, `incentives_mean`, `H3_num` (willingness to change)

x_label	y_label	n	rho	p	p_adj
Mean training need for data sharing	Mean barriers to sharing/publishing	50	0.42	0.00	0.03
Mean training need for data sharing	Mean motivating incentives	50	0.39	0.00	0.03
Mean training need for data sharing	Current open-science uptake (proportion)	50	-0.23	0.11	0.33

x_label	y_label	n	rho	p	p_adj
Mean training need for data sharing	Current file-drawer-targeted practices (count)	50	0.23	0.11	0.33
Methods/statistics courses completed	Mean barriers to sharing/publishing	50	0.18	0.22	0.52
Mean training need for data sharing	Ever shared data	49	-0.15	0.29	0.59
Methods/statistics courses completed	Mean motivating incentives	50	0.13	0.38	0.64
Methods/statistics courses completed	Current file-drawer-targeted practices (count)	50	-0.09	0.52	0.78
Methods/statistics courses completed	Ever shared data	49	-0.03	0.85	0.95
Methods/statistics courses completed	Current open-science uptake (proportion)	50	-0.03	0.85	0.95
Methods/statistics courses completed	Willingness to change practices	50	-0.02	0.87	0.95

x_label	y_label	n	rho	p	p_adj
Mean training need for data sharing	Willingness to change practices	50	-0.00	0.98	0.98

3) Awareness/implications (I2, I4) vs uptake and norms (I5)

Goal: test whether awareness of publication bias and perceived consequences connect to behavior and norms.

- Predictors (`x_vars`):
  - `I2_num` (self-rated familiarity with the file drawer problem)
  - `I4_severity_mean` and `I4_familiarity_mean` (means across the 7 I4 consequences)
- Outcomes (`y_vars`):
  - uptake: `open_science_present_prop`, `fd_practices_present_yes`
  - barriers/incentives: `barriers_mean`, `incentives_mean`
  - norms proxy: `qrp_unacceptability_mean` (mean of selected I5 QRP ratings; higher = more "unacceptable")

x_label	y_label	n	rho	p	p_adj
Mean familiarity with consequences	Current open-science uptake (proportion)	50	0.33	0.02	0.27
Mean familiarity with consequences	Mean barriers to sharing/publishing	50	-0.28	0.05	0.35
Mean familiarity with consequences	Current file-drawer-targeted practices (count)	50	0.23	0.10	0.49
Familiarity with the file drawer problem	Mean barriers to sharing/publishing	46	-0.21	0.16	0.49
Mean perceived severity of consequences	Mean motivating incentives	50	0.19	0.19	0.49

x_label	y_label	n	rho	p	p_adj
Familiarity with the file drawer problem	Mean unacceptability of questionable research practices	46	0.18	0.22	0.49
Mean perceived severity of consequences	Current file-drawer-targeted practices (count)	50	-0.17	0.23	0.49
Familiarity with the file drawer problem	Mean motivating incentives	46	0.17	0.27	0.50
Familiarity with the file drawer problem	Current open-science uptake (proportion)	46	0.16	0.30	0.50
Mean perceived severity of consequences	Mean barriers to sharing/publishing	50	0.12	0.40	0.61
Mean familiarity with consequences	Mean motivating incentives	50	0.10	0.49	0.67
Mean perceived severity of consequences	Current open-science uptake (proportion)	50	-0.08	0.57	0.72
Mean perceived severity of consequences	Mean unacceptability of questionable research practices	50	0.06	0.69	0.79

x_label	y_label	n	rho	p	p_adj
Mean familiarity with consequences	Mean unacceptability of questionable research practices	50	0.01	0.93	0.99
Familiarity with the file drawer problem	Current file-drawer-targeted practices (count)	46	0.00	0.99	0.99

4) Attrition experience (R1) vs awareness and uptake

Goal: test whether people with more file-drawer exposure differ in awareness, adoption, or perceived barriers.

- Predictors (`x_vars`): four complementary summaries of R1 stage counts:
  - `attrition_total_n` (raw count)
  - `attrition_share` (normalized)
  - `attrition_pre_sub_n` and `attrition_post_sub_n` (where attrition occurs)
- Outcomes (`y_vars`): awareness (`I2_num`, `I4_severity_mean`), uptake (`open_science_present_prop`, `fd_practices_present_yes`), and barriers/incentives (`barriers_mean`, `incentives_mean`)

x_label	y_label	n	rho	p	p_adj
Attrition before submission	Current file-drawer-targeted practices (count)	50	0.43	0.00	0.02
Total attrition	Current file-drawer-targeted practices (count)	50	0.43	0.00	0.02
Attrition share = attrition/(attrition+published)	Mean barriers to sharing/publishing	46	0.34	0.02	0.18

x_label	y_label	n	rho	p	p_adj
Attrition at submission/review	Mean barriers to sharing/publishing	50	0.27	0.06	0.38
Attrition share = attrition/(attrition+published)	Mean perceived severity of consequences	46	-0.25	0.09	0.43
Total attrition	Current open-science uptake (proportion)	50	0.19	0.18	0.68
Attrition share = attrition/(attrition+published)	Current file-drawer-targeted practices (count)	46	0.18	0.23	0.68
Attrition before submission	Current open-science uptake (proportion)	50	0.18	0.21	0.68
Total attrition	Mean barriers to sharing/publishing	50	0.16	0.26	0.68
Attrition at submission/review	Current file-drawer-targeted practices (count)	50	0.15	0.28	0.68
Attrition at submission/review	Mean perceived severity of consequences	50	0.14	0.32	0.71

x_label	y_label	n	rho	p	p_adj
Attrition at submission/review	Familiarity with the file drawer problem	46	-0.12	0.43	0.87
Attrition at submission/review	Current open-science uptake (proportion)	50	0.10	0.49	0.87
Attrition at submission/review	Mean motivating incentives	50	-0.09	0.54	0.87
Attrition before submission	Mean barriers to sharing/publishing	50	0.08	0.57	0.87
Attrition share = attrition/(attrition+published)	Mean motivating incentives	46	-0.08	0.61	0.87
Attrition share = attrition/(attrition+published)	Familiarity with the file drawer problem	42	-0.08	0.63	0.87
Attrition share = attrition/(attrition+published)	Current open-science uptake (proportion)	46	0.07	0.65	0.87
Attrition before submission	Mean perceived severity of consequences	50	-0.04	0.80	0.95

x_label	y_label	n	rho	p	p_adj
Total attrition	Mean perceived severity of consequences	50	-0.02	0.87	0.95
Total attrition	Familiarity with the file drawer problem	46	-0.02	0.91	0.95
Attrition before submission	Mean motivating incentives	50	0.01	0.93	0.95
Attrition before submission	Familiarity with the file drawer problem	46	0.01	0.95	0.95
Total attrition	Mean motivating incentives	50	-0.01	0.95	0.95

Broader screening (demographics → variables of interest)

Goal: a wide, pre-specified dredge: relate a small set of demographic/productivity predictors to *all* other derived variables.

- Predictors (`demo_vars`): years active, career stage, early-career indicator, methods courses, and publication/productivity counts.
- Outcomes (`outcome_vars`): everything else in `keep_vars` (attrition indices, awareness indices, uptake indices, etc.).

The output shows the top 30 associations ranked by BH-adjusted `p_adj` (within this wide screen), then by effect size.

x_label	y_label	n	rho	p	p_adj
Quantitative projects per year	Barrier subscale: time/incentives/pressure	50	-0.54	0.00	0.01

x_label	y_label	n	rho	p	p_adj
Projects per year	Current open-science uptake (proportion)	50	0.54	0.00	0.01
Projects per year	Published projects	50	0.52	0.00	0.01
Quantitative projects per year	Mean barriers to sharing/publishing	50	-0.50	0.00	0.01
Quantitative projects per year	Published projects	50	0.50	0.00	0.01
Peer-reviewed publications per year	Barrier subscale: time/incentives/pressure	50	-0.49	0.00	0.01
Peer-reviewed publications per year	Mean barriers to sharing/publishing	50	-0.47	0.00	0.01
Peer-reviewed publications per year	Published projects	50	0.44	0.00	0.03
Career stage (ordinal)	Mean perceived severity of consequences	49	-0.44	0.00	0.03
Projects per year	Future file-drawer-targeted practices (count)	50	0.41	0.00	0.05
Projects per year	Attrition before submission	50	0.41	0.00	0.05

x_label	y_label	n	rho	p	p_adj
Projects per year	Barrier subscale: time/incentives/pressure	50	-0.40	0.00	0.06
Projects per year	Ever shared data	49	0.39	0.01	0.07
Quantitative projects per year	Ever shared data	49	0.39	0.01	0.07
Quantitative projects per year	Current open-science uptake (proportion)	50	0.38	0.01	0.08
Early career (student/postdoc/assistant)	Mean perceived severity of consequences	49	0.37	0.01	0.09
Projects per year	Total attrition	50	0.37	0.01	0.09
Quantitative projects per year	Barrier subscale: skills/technical barriers	50	-0.37	0.01	0.09
Peer-reviewed publications per year	Barrier subscale: external restrictions/policy	50	-0.35	0.01	0.11
Projects per year	Mean barriers to sharing/publishing	50	-0.35	0.01	0.11

x_label	y_label	n	rho	p	p_adj
Peer-reviewed publications per year	Current open-science uptake (proportion)	50	0.33	0.02	0.15
Quantitative projects per year	Attrition before submission	50	0.33	0.02	0.15
Quantitative projects per year	Barrier subscale: external restrictions/policy	50	-0.33	0.02	0.16
Years active in experimental philosophy	Barrier subscale: skills/technical barriers	50	-0.32	0.02	0.16
Projects per year	Micropublications frequency (ordinal)	50	0.32	0.02	0.16
Projects per year	Barrier subscale: skills/technical barriers	50	-0.32	0.02	0.16
Projects per year	Current file-drawer-targeted practices (count)	50	0.31	0.03	0.18
Peer-reviewed publications per year	Barrier subscale: skills/technical barriers	50	-0.31	0.03	0.18

x_label	y_label	n	rho	p	p_adj
Peer-reviewed publications per year	Ever shared data	49	0.30	0.03	0.21
Methods/statistics courses completed	Mean perceived severity of consequences	50	0.30	0.04	0.22

Notes for the paper draft (sanity check against existing results)

What this does: recomputes the means and SDs of the six R1 stage-count variables directly from the dataset (no modeling), to cross-check numbers reported elsewhere in the draft.

Stage	M	SD
[[character]]	1.66	3.10
[[character]]	0.88	1.53
[[character]]	0.98	1.79
[[character]]	0.26	0.78
[[character]]	0.38	0.78
[[character]]	6.98	9.66

Extended EDA modules (dredging on purpose)

5) Latent "barrier types" (S5) and practical capacity (H4)

What analysis is this? Exploratory factor analysis (EFA) on the barrier/training item set.

- Inputs: 11 S5 barrier items (1–5) + 5 H4 training-need items (1–5), treated as numeric/ordinal scores.
- Step 1 (choose number of factors): `psych::fa.parallel()` (parallel analysis) suggests a factor count (`nf`).
- Step 2 (fit factor model): `psych::fa(..., fm = "minres", rotate = "oblimin")`.
  - `minres` = minimum residual factoring (common in EFA).
  - `oblimin` = oblique rotation (allows factors to correlate, which is plausible here).
- Output 1: a loadings table (how strongly each item loads on each latent factor).
- Output 2: factor scores (one score per respondent per factor), then correlations of those scores with outcomes (`data_shared_ever`, `open_science_*`, `attrition_share`).

```
## Parallel analysis suggests that the number of factors =  3  and the number of components =  NA
```

item	MR1	MR3	MR2
IP/ownership concerns	-0.01	0.01	0.84
Fear of misuse/misrepresentation	0.21	-0.02	0.68
Competitive advantage concerns	0.04	0.09	0.71
Lack of direct benefits/incentives	-0.11	0.89	-0.05
Technical challenges	0.12	0.20	-0.01
Journal policies don’t require sharing	0.10	0.57	0.06
Fear of criticism/exposing errors	0.04	0.46	0.26
Pressure to publish in high-impact journals	-0.04	0.63	0.34
External restrictions (funders/partners)	-0.17	0.07	0.57
Lack of training/skills	0.24	0.54	0.06

item	MR1	MR3	MR2
Extra effort feels overwhelming	0.26	0.71	-0.04
Make data useful to others	0.87	0.17	-0.15
Platforms/repositories/tools	0.74	-0.02	0.18
Protect confidentiality	0.68	-0.05	0.04
Reuse rights (copyright/licensing/citations)	0.68	-0.12	0.19
Long-term preservation/archiving	0.70	0.01	-0.00

x_label	y_label	n	rho	p	p_adj
S5/H4 latent factor 3	Ever shared data	49	-0.48	0.00	0.01
S5/H4 latent factor 3	Attrition share = attrition/(attrition+published)	46	0.41	0.00	0.02
S5/H4 latent factor 2	Current open-science uptake (proportion)	50	-0.39	0.01	0.02

x_label	y_label	n	rho	p	p_adj
S5/H4 latent factor 2	Future open-science uptake (proportion)	50	-0.30	0.04	0.11
S5/H4 latent factor 1	Current open-science uptake (proportion)	50	-0.25	0.08	0.17
S5/H4 latent factor 2	Ever shared data	49	-0.25	0.09	0.17
S5/H4 latent factor 2	Attrition share = attrition/(attrition+published)	46	0.22	0.14	0.20
S5/H4 latent factor 1	Attrition share = attrition/(attrition+published)	46	0.21	0.15	0.20
S5/H4 latent factor 3	Future open-science uptake (proportion)	50	-0.21	0.15	0.20
S5/H4 latent factor 1	Future open-science uptake (proportion)	50	-0.20	0.16	0.20
S5/H4 latent factor 1	Ever shared data	49	-0.11	0.45	0.49

x_label	y_label	n	rho	p	p_adj
S5/H4 latent factor 3	Current open-science uptake (proportion)	50	-0.04	0.81	0.81

5a) Theory-driven subscales (interpretable indices)

What analysis is this? Hand-built, interpretable indices constructed as means of specific S5 items.

Each subscale is a respondent-level mean of the listed items (all on the same 1–5 scale):

- `barriers_defensive` = mean of S5[SQ001] (IP/ownership), S5[SQ002] (misuse fears), S5[SQ003] (competition)
- `barriers_skills` = mean of S5[SQ005] (technical challenges), S5[SQ010] (lack of training/skills)
- `barriers_time_incentives` = mean of S5[SQ004] (lack of incentives), S5[SQ008] (high-impact pressure), S5[SQ011] (overwhelming effort)
- `barriers_external_policy` = mean of S5[SQ006] (journal policy), S5[SQ009] (external restrictions)

We then correlate these subscales with key outcomes (data sharing, uptake, attrition share).

x_label	y_label	n	rho	p	p_adj
Barrier subscale: defensive concerns (IP/misuse/competition)	Ever shared data	49	-0.53	0.00	0.00
Barrier subscale: skills/technical barriers	Current open-science uptake (proportion)	50	-0.43	0.00	0.01
Barrier subscale: external restrictions/policy	Attrition share = attrition/(attrition+published)	46	0.37	0.01	0.06
Barrier subscale: skills/technical barriers	Ever shared data	49	-0.34	0.02	0.06

x_label	y_label	n	rho	p	p_adj
Barrier subscale: time/incentives/pressure	Current open-science uptake (proportion)	50	-0.32	0.02	0.07
Barrier subscale: defensive concerns (IP/misuse/competition)	Attrition share = attrition/(attrition+published)	46	0.33	0.03	0.07
Barrier subscale: time/incentives/pressure	Future open-science uptake (proportion)	50	-0.26	0.07	0.16
Barrier subscale: time/incentives/pressure	Attrition share = attrition/(attrition+published)	46	0.22	0.14	0.24
Barrier subscale: defensive concerns (IP/misuse/competition)	Future open-science uptake (proportion)	50	-0.22	0.12	0.24
Barrier subscale: external restrictions/policy	Ever shared data	49	-0.21	0.15	0.24
Barrier subscale: time/incentives/pressure	Ever shared data	49	-0.20	0.17	0.24

x_label	y_label	n	rho	p	p_adj
Barrier subscale: skills/technical barriers	Future open-science uptake (proportion)	50	-0.19	0.18	0.25
Barrier subscale: skills/technical barriers	Attrition share = attrition/(attrition+published)	46	0.18	0.22	0.27
Barrier subscale: external restrictions/policy	Current open-science uptake (proportion)	50	-0.05	0.72	0.77
Barrier subscale: external restrictions/policy	Future open-science uptake (proportion)	50	0.05	0.72	0.77
Barrier subscale: defensive concerns (IP/misuse/competition)	Current open-science uptake (proportion)	50	-0.04	0.78	0.78

6) Practice ecology: which open-science behaviors "bundle" together? (S4)

What analysis is this? Two complementary "behavior ecology" views of S4 *present* practices:

1) **Co-occurrence matrix**: phi correlations (Pearson correlations on 0/1 binaries) between every pair of S4 items.

2) **Rule-style summaries**: for each ordered pair A→B, compute support/confidence/lift (association-rule metrics).

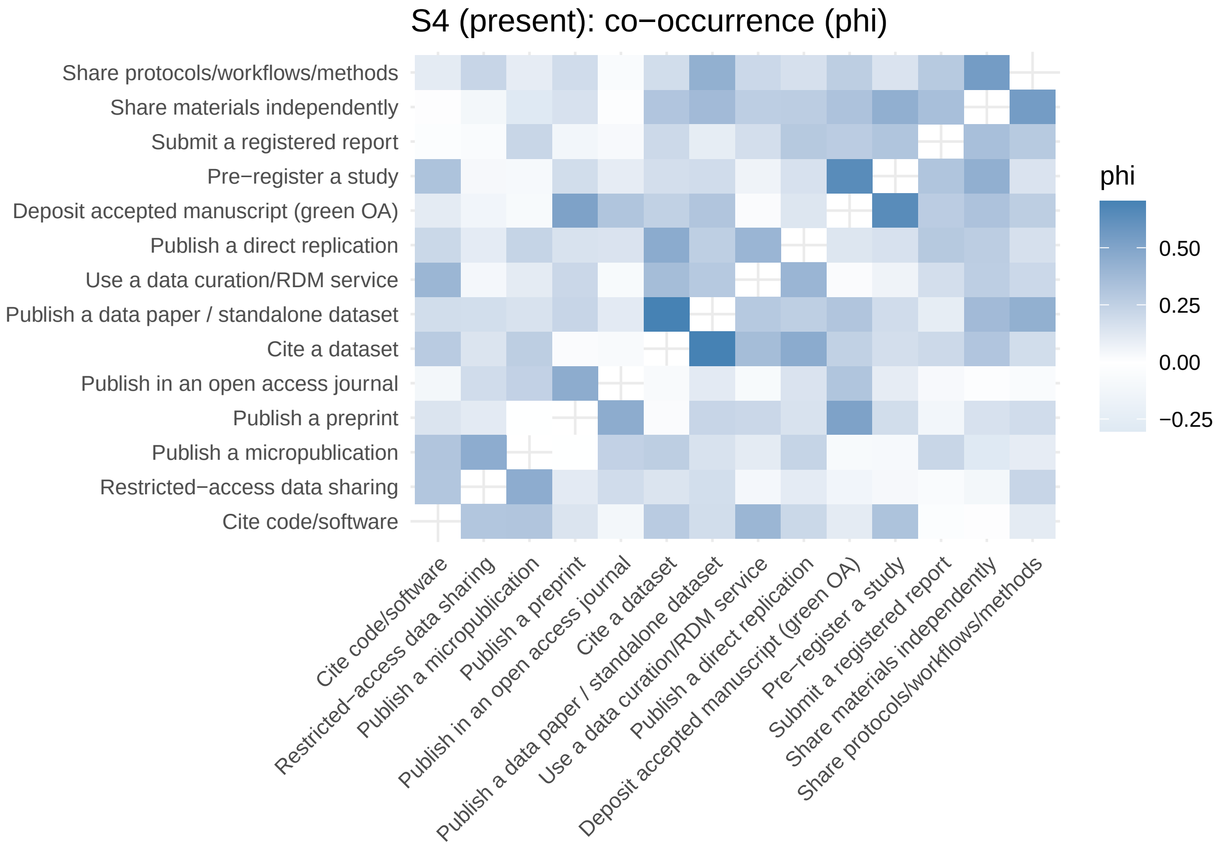
6a) Pairwise co-occurrence (phi correlations) + clustering

Inputs: `s4p` is the respondent-by-item matrix of S4 present items, recoded:

- Yes = 1, No = 0, I don't know = missing

Statistic: `phi_mat = cor(s4p, use = "pairwise.complete.obs")`. Because items are binary, this is a phi correlation.

Heatmap ordering: hierarchical clustering on distance `(1 - phi)/2` is used *only* to make visual blocks easier to see.



label_x	label_y	phi

6b) "Gateway" rules (support / confidence / lift)

Goal: identify potentially "gateway-like" relationships among practices (purely descriptive).

For each ordered pair A→B (two S4 items), we compute:

- support = P(A = 1 and B = 1)
- confidence = P(B = 1 | A = 1)
- lift = confidence / P(B = 1)

We restrict to rules with support   0.10 to avoid extremely rare co-occurrences.

A	B	n	support	confidence	lift
[[character]]	[[character]]	39	0.26	1.00	2.44
[[character]]	[[character]]	39	0.26	0.62	2.44
[[character]]	[[character]]	45	0.18	0.80	2.00

A	B	n	support	confi-dence	lift
[[charac-ter]]	[[charac-ter]]	45	0.18	0.44	2.00
[[charac-ter]]	[[charac-ter]]	36	0.14	0.71	1.98
[[charac-ter]]	[[charac-ter]]	36	0.14	0.38	1.98
[[charac-ter]]	[[charac-ter]]	43	0.16	0.78	1.97
[[charac-ter]]	[[charac-ter]]	43	0.16	0.41	1.97
[[charac-ter]]	[[charac-ter]]	41	0.12	0.62	1.83
[[charac-ter]]	[[charac-ter]]	41	0.12	0.36	1.83
[[charac-ter]]	[[charac-ter]]	39	0.26	0.62	1.74
[[charac-ter]]	[[charac-ter]]	39	0.13	0.31	1.74
[[charac-ter]]	[[charac-ter]]	39	0.26	0.71	1.74
[[charac-ter]]	[[charac-ter]]	39	0.13	0.71	1.74
[[charac-ter]]	[[charac-ter]]	40	0.17	0.33	1.67
[[charac-ter]]	[[charac-ter]]	40	0.17	0.87	1.67
[[charac-ter]]	[[charac-ter]]	39	0.21	1.00	1.62
[[charac-ter]]	[[charac-ter]]	39	0.21	0.33	1.62
[[charac-ter]]	[[charac-ter]]	45	0.36	0.89	1.60
[[charac-ter]]	[[charac-ter]]	45	0.36	0.64	1.60

7) Stability-based dredging (bootstrap "which predictors keep showing up?")

What analysis is this? A nonparametric "stability" check for correlation-based dredging.

For a chosen outcome (e.g., `data_shared_ever`), we:

1) Compute Spearman correlations between the outcome and each predictor (full sample) → `rho`.
2) Repeat B times (bootstrap resampling respondents with replacement):
    - recompute the correlations,

- rank predictors by |rho|,
- keep the top `top_k` predictors.

3) For each predictor, compute `pick_freq` = proportion of bootstrap runs where it appears in the top `top_k`.

Interpretation: high `pick_freq` means the predictor is *consistently among the strongest correlates* under resampling (still exploratory, not causal).

outcome	pred	pick_freq	rho
Attrition share = attrition/(attrition+published)	Barrier subscale: external restrictions/policy	0.75	0.37
Attrition share = attrition/(attrition+published)	Mean barriers to sharing/publishing	0.64	0.34
Attrition share = attrition/(attrition+published)	Barrier subscale: defensive concerns (IP/misuse/competition)	0.53	0.33
Attrition share = attrition/(attrition+published)	Methods/statistics courses completed	0.50	0.30
Attrition share = attrition/(attrition+published)	Mean perceived severity of consequences	0.39	-0.25
Attrition share = attrition/(attrition+published)	Years active in experimental philosophy	0.36	-0.25

outcome	pred	pick_freq	rho
Attrition share = attrition/(attrition+published)	Mean training need for data sharing	0.25	0.22
Attrition share = attrition/(attrition+published)	Barrier subscale: time/incentives/pressure	0.24	0.22
Attrition share = attrition/(attrition+published)	Peer-reviewed publications per year	0.23	-0.20
Attrition share = attrition/(attrition+published)	Quantitative projects per year	0.18	-0.20
Ever shared data	Barrier subscale: defensive concerns (IP/misuse/competition)	0.99	-0.53
Ever shared data	Projects per year	0.78	0.39
Ever shared data	Quantitative projects per year	0.75	0.39
Ever shared data	Mean barriers to sharing/publishing	0.72	-0.39

outcome	pred	pick_freq	rho
Ever shared data	Barrier subscale: skills/technical barriers	0.53	-0.34
Ever shared data	Peer-reviewed publications per year	0.31	0.30
Ever shared data	Sponsor requirements for open dissemination (count)	0.28	0.23
Ever shared data	Mean motivating incentives	0.15	-0.25
Ever shared data	Years active in experimental philosophy	0.12	0.12
Ever shared data	Mean familiarity with consequences	0.07	-0.07
Current open-science uptake (proportion)	Projects per year	0.96	0.54
Current open-science uptake (proportion)	Barrier subscale: skills/technical barriers	0.78	-0.43

outcome	pred	pick_freq	rho
Current open-science uptake (proportion)	Quantitative projects per year	0.59	0.38
Current open-science uptake (proportion)	Mean familiarity with consequences	0.42	0.33
Current open-science uptake (proportion)	Peer-reviewed publications per year	0.39	0.33
Current open-science uptake (proportion)	Mean barriers to sharing/publishing	0.36	-0.33
Current open-science uptake (proportion)	Barrier subscale: time/incentives/pressure	0.33	-0.32
Current open-science uptake (proportion)	Sponsor requirements for open dissemination (count)	0.27	0.25
Current open-science uptake (proportion)	Career stage (ordinal)	0.22	0.23

outcome	pred	pick_freq	rho
Current open-science uptake (proportion)	Mean unacceptability of questionable research practices	0.21	0.21

8) File-drawer pathway typology (R1 + R2–R6)

What analysis is this? A crude typology of "why projects stop" among respondents who report any attrition.

- We first restrict to respondents with `attrition_total_n > 0`.
- For the stage-specific reason items (R2–R6), responses are strings like `"Yes"`, `"No"`, and `"N/A"`.
  - We recode `"Yes"` → 1 and everything else → 0 (treating `"N/A"` as "not selected").
- We then map each reason item into a coarse category:
  - results-independent (resources/interest; more studies needed)
  - quality/concerns
  - results-dependent (null; unimportant/unanticipated; unimportant by editor/reviewers)
  - structural (journal fit/scope)
- For each respondent, we count how many items they selected in each category and label the category with the highest count as `dominant`.

This is intentionally reductive (multi-select reasons are collapsed into a single dominant label) and should be treated as a descriptive device.

dominant	n
Quality/concerns	21
Results-independent (more studies needed)	6
Results-dependent (null)	5
Results-independent (resources/interest)	3
Results-dependent (unimportant/unanticipated)	2

dominant	n	open_science_present_prop	data_shared_ever	attrition_total_n
Quality/concerns	21	0.49	0.90	6.67
Results-independent (more studies needed)	6	0.44	0.83	4.67
Results-dependent (null)	5	0.47	0.80	3.20
Results-independent (resources/interest)	3	0.44	0.67	5.33
Results-dependent (unimportant/unanticipated)	2	0.54	1.00	4.00

9) Open text dredging (lightweight regex coding)

What analysis is this? A minimal "keyword flagging" pass over the open-text fields.

- We concatenate multiple open text columns (S3 comments, S5 "Other", venues, final comments, etc.) into one `text_all` string per row in the public *text-only* dataset.
- We then create binary flags for platform mentions (e.g., OSF) and theme keywords (e.g., "time/effort") using regex patterns.
- Outputs:
  - a prevalence table (how many respondents mention each keyword/theme)
  - (no correlations in the public report; see note below)

Because mentions are sparse, many flags will have near-zero variance; treat this as a coarse exploratory supplement, not a robust text analysis.

flag	n	pct
mention_philarchive	19.00	38.00
mention_arxiv	19.00	38.00
mention_osf	15.00	30.00

flag	n	pct
theme_time_effort	5.00	10.00
theme_journals_review	4.00	8.00
mention_ssrn	3.00	6.00
mention_researchgate	3.00	6.00
theme_incentives	3.00	6.00
mention_github	1.00	2.00
mention_jasnh	1.00	2.00
theme_fear_criticism	1.00	2.00
mention_academia	0.00	0.00
mention_zenodo_figshare	0.00	0.00
theme_ip_misuse	0.00	0.00

Note. In the public release, open-text responses are distributed as a separate, **unlinked** dataset; therefore, we do not compute respondent-level correlations between text flags and the main survey variables here.